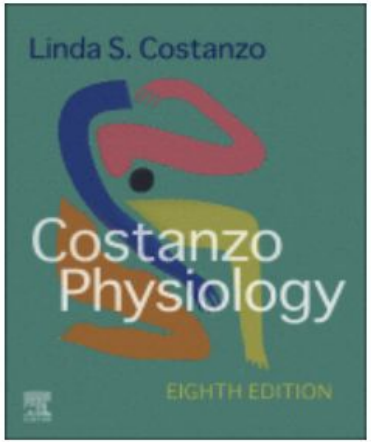
PHYSIOLOGY:
Body Fluid Compartments, Volume Expansions and Contractions

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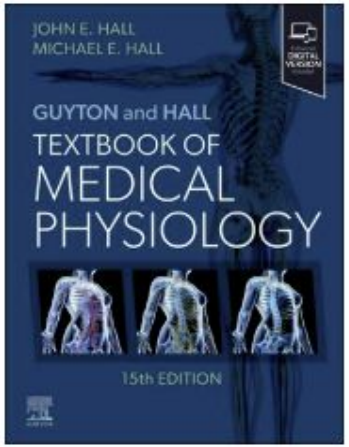


RENL Fall 2026

Resources



Ch. 6



Ch. 25



OSMOSIS: Movement of water between body compartments

***** You may upload these lecture materials into Gemini/NotebookLM.***

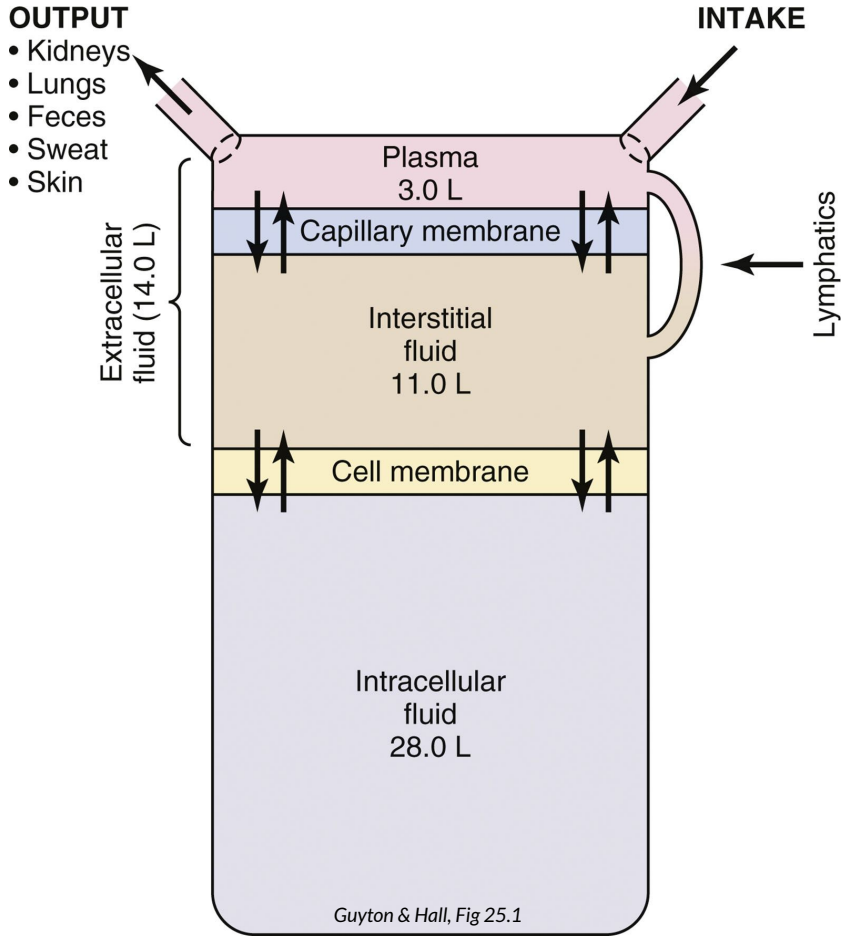
Session Objectives

1. **Describe** the distribution of total body water among intracellular, extracellular, plasma, and interstitial compartments.
2. **Calculate** the osmolarity of simple solutions and explain how tonicity determines water movement and cell volume.
3. **Interpret** Darrow–Yannet diagrams to predict ECF volume, ICF volume, and osmolarity after equilibration in the six acute fluid disturbances.
4. **Explain** the changes in RBC volume, hematocrit, and plasma protein concentration associated with each disturbance.

What are the body fluid compartments?

Total Body Water (TBW) ≈ **60%** of body weight

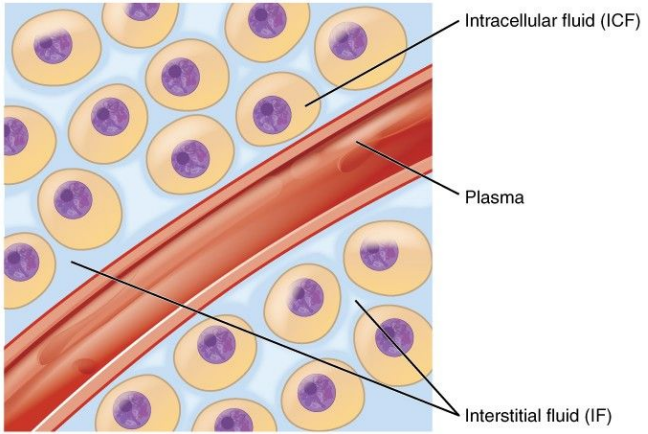
- **Intracellular fluid (ICF)** ≈ **40%** of body weight; **⅔ TBW**
- **Extracellular fluid (ECF)** ≈ **20%** of body weight; **⅓ TBW**
  - **Plasma** (¼ of ECF) = fluid circulating in the blood vessels; contains proteins that are largely retained by the capillary wall
  - **Interstitial fluid** (¾ of ECF) = fluid that bathes cells; forms when plasma is filtered across capillary wall; contains little protein

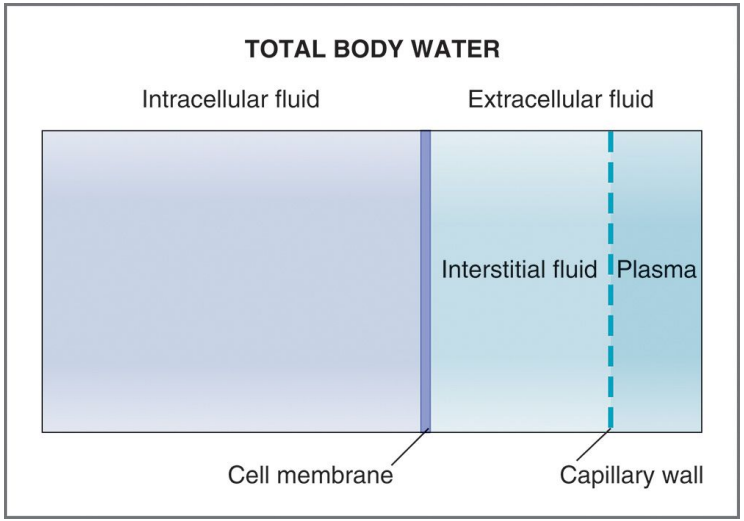


Two barriers explain most fluid distribution

- **Cell membrane:** separates ICF from ECF; water crosses readily, while many solutes are restricted.
- **Capillary wall:** separates plasma from interstitial fluid; water and small solutes cross much more readily than plasma proteins.

Blood plasma and interstitial fluid exchange water, ions, and small solutes freely across capillary walls, making their ionic compositions nearly identical, with one major exception: **plasma contains a much higher concentration of proteins.**





Review: Solute Concentration, Osmolarity, Tonicity

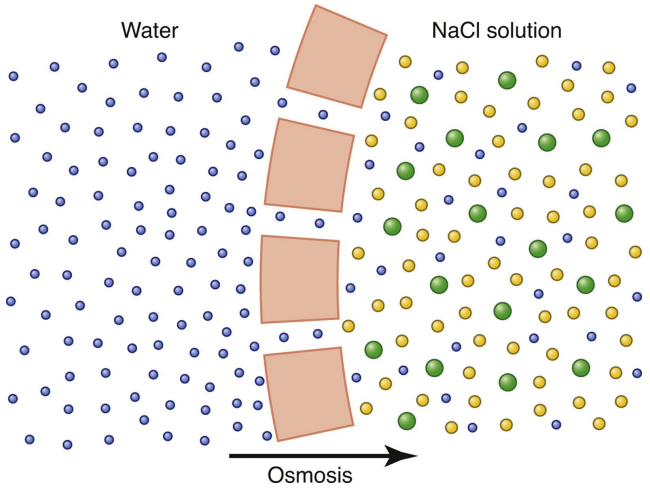
- **Moles** = amount of substance, 1 mole = 6×10^{23} entities (e.g., molecules, ions)
- **Molarity** = solute concentration; mmol/L = mM
- **Osmoles** = # of osmotically active particles
- # of particles a solute dissociates into = **g**
- **Osmolarity = g × C** (g = particles per molecule; C = concentration)
  - ***What is the osmolarity of a 150 mM solution of NaCl?***
    - NaCl dissociates into Na^+ and Cl^- so **g=2.**
    - Osmolarity = g x C = (2) x (150 mmol/L) = 300 mOsm/L
  - ***What is the osmolarity of a 150 mM solution of glucose?***
- **Osmolality:** osmoles/kg of water
- Normal plasma osmolality: approximately 275–295 mOsm/kg. For our diagrams, use a simplified baseline osmolarity of 300 mOsm/L.

Review: Solute Concentration, Osmolarity, Tonicity *(cont')*

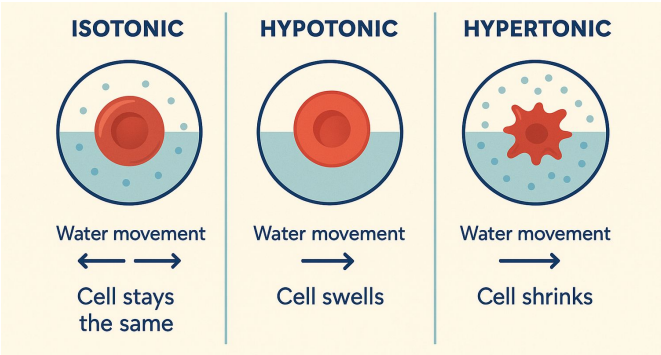
- **Osmosis:** water flows across a semipermeable membrane **toward higher <u>osmolarity</u>**.
- **Osmotic pressure** = how strongly a solution "pulls" water towards it.

Which of the following solutions would exert the greatest osmotic pressure?

A. 300 mM glucose
B. 300 mM NaCl
C. 300 mM $CaCl_2$

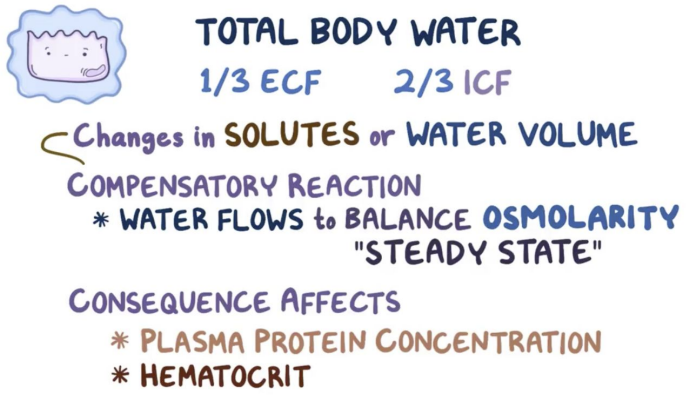


- **Tonicity** is the ability of an extracellular solution to make water move into or out of a cell by osmosis.



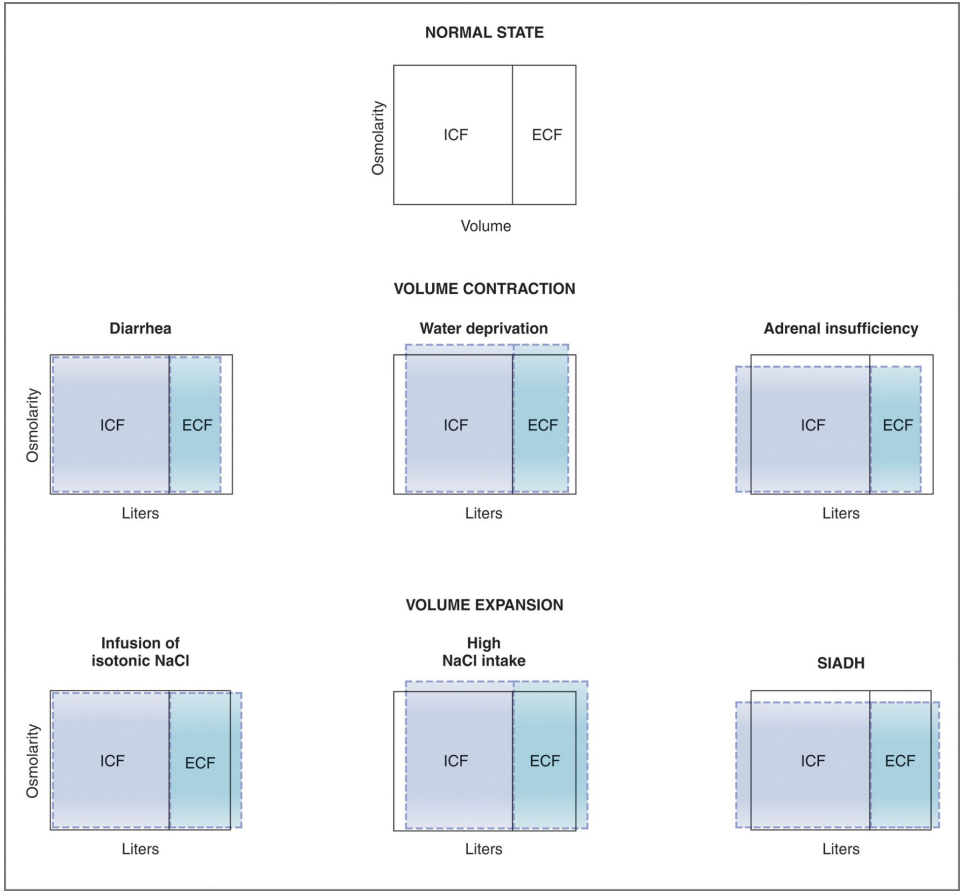
Events/conditions can cause acute fluid shifts between compartments.

- **"Events/conditions"** = changes in solute concentration or water volume <u>in the ECF</u>
- If the event/condition changes the osmolarity of the ECF, then a fluid shift will occur between the ECF and ICF.
- Net water movement stops once the ECF and ICF are at the same osmolarity; a "new equilibrium" will have been reached.
- If the event/condition does not change the osmolarity of the ECF, then no fluid shift will occur. *(How come?)*
- For each event/condition, consider the consequence on:
  - ECF volume
  - ICF volume
  - Osmolarity after equilibration
  - Hematocrit
  - Plasma protein concentration



How water can shift between compartments

- Shifts are characterized by changes in **ECF volume and osmolarity**.
  - ECF *volume*:
    - ↑ = “expansion”
    - ↓ = “contraction”
  - ECF *osmolarity*:
    - ↑ = “hyperosmotic”
    - ↓ = “hyposmotic”
    - No change = “isosmotic”
- 6 possible combinations (all refer to ECF):
  - Hyperosmotic, hyposmotic or isosmotic <u>expansion</u>
  - Hyperosmotic, hyposmotic or isosmotic <u>contraction</u>

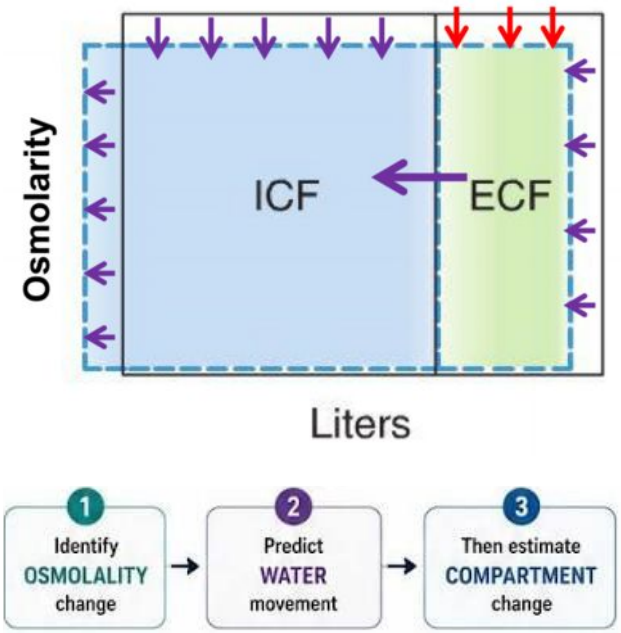


Rules for Acute Fluid Shifts

1. **Identify the event/condition.**
2. **Determine how the event/condition will impact <u>ECF</u> volume and osmolarity.** *Depends on nature of the event/condition (e.g., drinking a gallon of water vs. eating a large bag of salty potato chips). Is the ECF now hyperosmotic, hyposmotic or isosmotic compared to the ICF?*
3. **Water will move to the compartment with higher osmolarity (either ICF or ECF) and will stop moving when osmotic equilibrium is restored.** *Describe water movement in terms of ECF volume: did it contract or expand?*
4. **Assume that the # of osmoles in ICF and ECF remains constant unless the event changes them.** *Na^+ remains extracellular, and intracellular osmoles are held constant.*

How to Read a Darrow-Yannet Diagram

- Width = compartment volume
- Height = osmolarity
- Area = quantity of osmoles
- Baseline: ICF 28 L; ECF 14 L; both 300 mOsm/L
- Solid outline = baseline; dashed outline = final state (after equilibration)
- Always put ICF on left and ECF on right.
- Note that ICF is typically ⅔ TBW, and ECF is ⅓ TBW.
- For these examples, the relevant solutes are treated as restricted to their compartments.

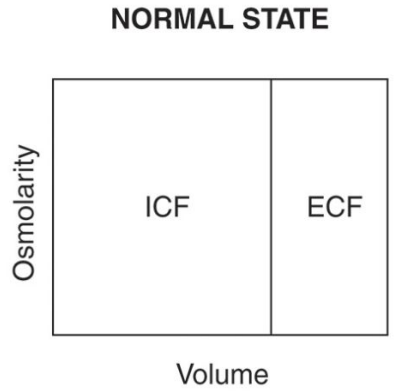


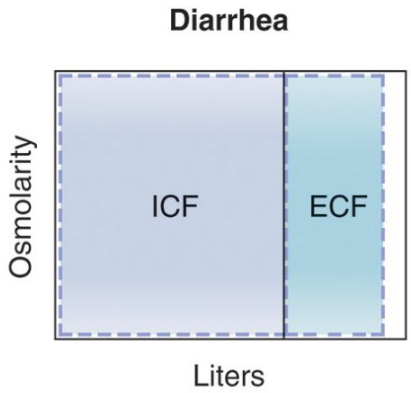
After equilibration, the height of the ICF box always equals the height of the ECF box. Why?

1) Isosmotic Volume Contraction

- **Event:** Diarrhea with loss of approx. isotonic fluid
- **What happens:** ECF volume decreases; osmolarity of remaining ECF is unchanged.
- **Fluid shift:** No net water movement between ECF and ICF
- **Compartments: ECF volume** ↓, osmolarity unchanged, ICF volume unchanged
- **Blood findings:** RBC volume unchanged; **hematocrit** ↑; **plasma protein concentration** ↑ *(Recall that all 3 of these are potentially impacted by ECF changes. How?)*

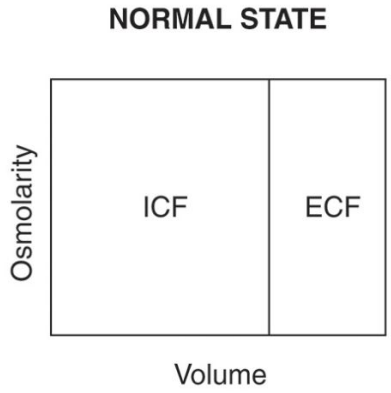
Why are hematocrit and plasma protein concentration increased?

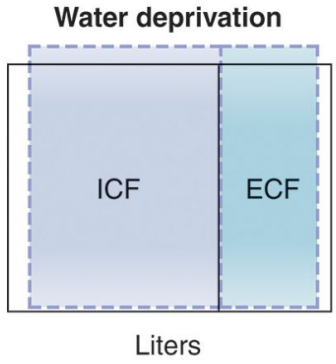




2) Hyperosmotic Volume Contraction

- **Event:** Unreplaced hypotonic sweat loss (e.g., sweating, fever); dilute polyuria (diabetes insipidus); water deprivation
- **What happens:** ECF volume decreases; osmolarity of remaining ECF increases. Water shifts from ICF to ECF.
- **Fluid shift:** From ICF to ECF
- **Compartments: ECF volume ↓, osmolarity↑, ICF volume ↓**
- **Blood findings: RBC volume ↓**; hematocrit approx. unchanged; **plasma protein concentration ↑**



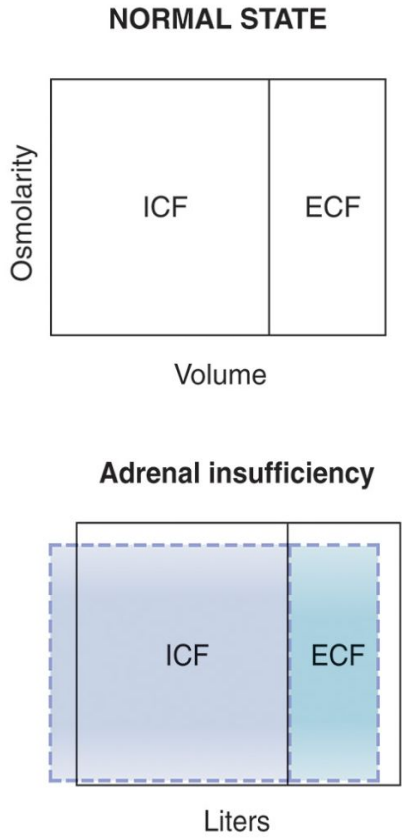


- ***Decreased plasma volume would be expected to increase hematocrit. However, increased osmolarity causes RBCs to shrink. These findings offset each other, resulting in no apparent hematocrit change.***
- ***However, this offset is not seen in all clinical scenarios.***

3) Hyposmotic Volume Contraction

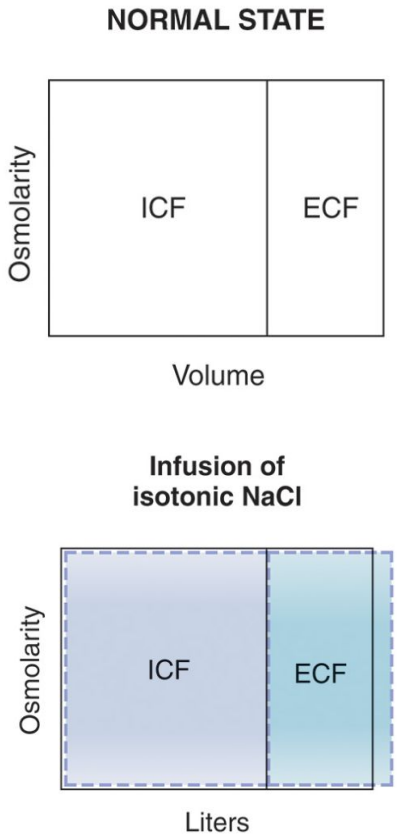
- **Event:** Renal salt loss from aldosterone deficiency in **primary adrenal insufficiency**
- **What happens:** *Aldosterone normally increases Na^+ reabsorption in the nephron, and water passively follows. Insufficient aldosterone leads to increased excretion of <u>both</u> Na^+ and water.* ECF volume decreases; osmolarity of remaining ECF decreases, water shifts from ECF to ICF. Why?
- **Fluid shift:** From ECF to ICF
- **Compartments: ECF volume ↓, osmolarity↓, ICF volume ↑**
- **Blood findings: RBC volume ↑; hematocrit ↑; plasma protein concentration ↑**

TIP: Always use the <u>height</u> of the ECF box (i.e., osmolarity) to determine if RBC volume will increase, decrease or remain unchanged.



4) Isosmotic Volume Expansion

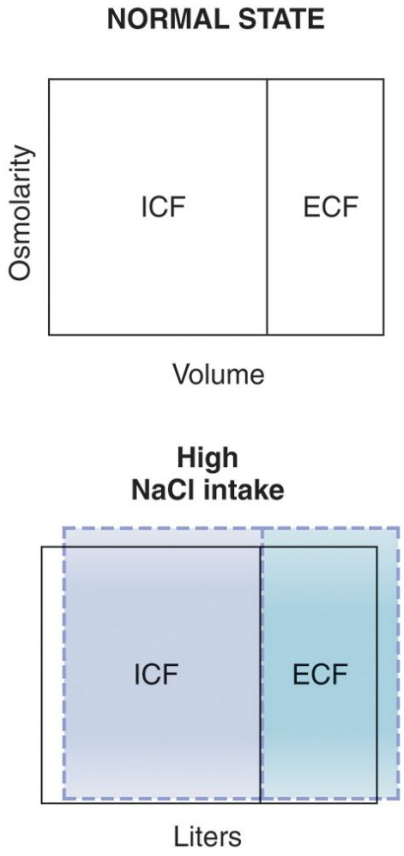
- **Event:** Infusion of isotonic NaCl (0.9% saline)
- **What happens:** ECF volume increases; osmolarity of remaining ECF is unchanged.
- **Fluid shift:** No net water movement between ECF and ICF
- **Compartments: ECF volume** ↑; osmolarity unchanged; ICF volume unchanged
- **Blood findings:** RBC volume unchanged; **hematocrit** ↓; **plasma protein concentration** ↓



TIP: Use the <u>width</u> of the ECF box (i.e., volume) to determine if plasma protein concentration will increase, decrease or remain unchanged.

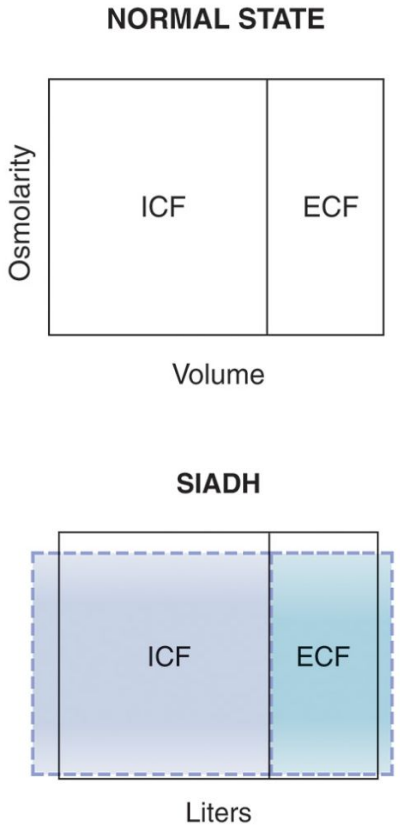
5) Hyperosmotic Volume Expansion

- **Event:** High NaCl intake
- **What happens:** Added NaCl increases ECF osmolarity, drawing water from ICF to ECF and expanding ECF volume.
- **Fluid shift:** From ICF to ECF
- **Compartments: ECF volume ↑, osmolarity↑, ICF volume ↓**
- **Blood findings: RBC volume ↓; hematocrit ↓, plasma protein concentration ↓**

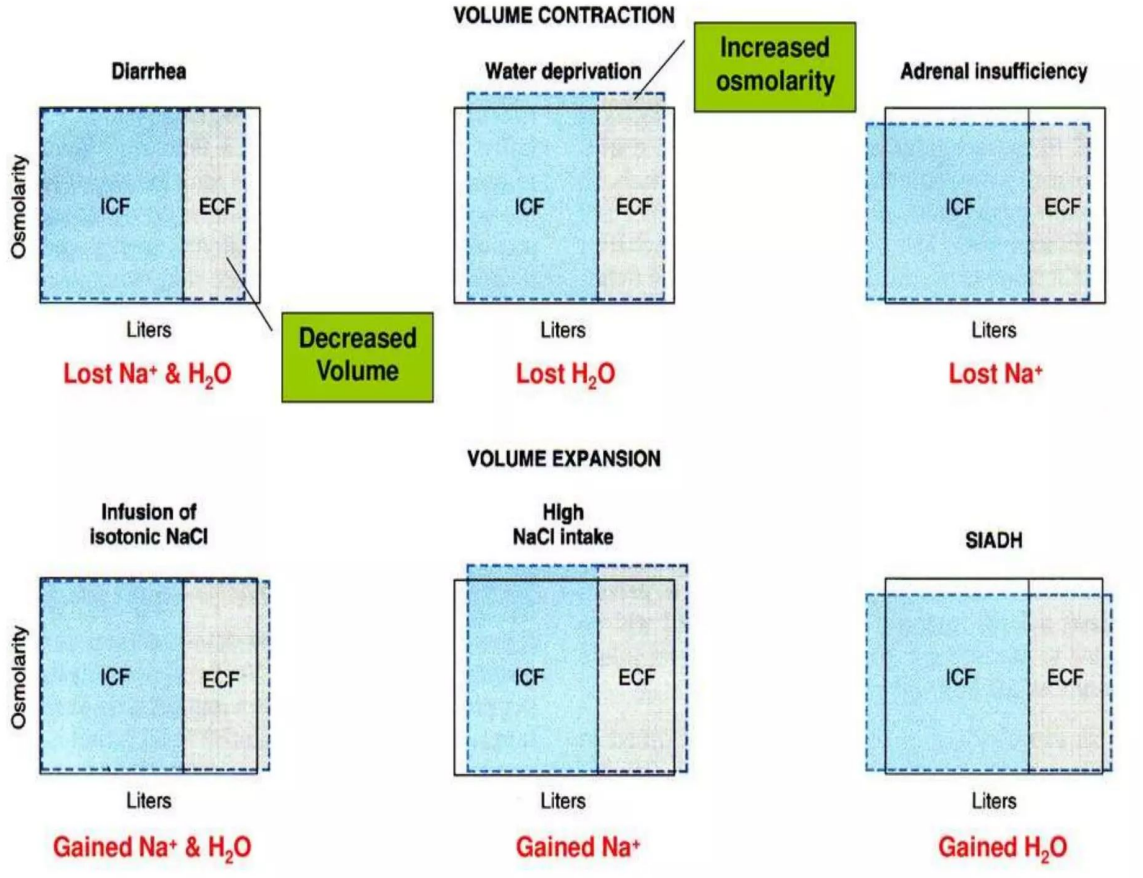


6) Hyposmotic Volume Expansion

- **Event:** Syndrome of inappropriate (excessive) antidiuretic hormone secretion (SIADH); water intake exceeding renal excretion
- **What happens:** *ADH normally increases water reabsorption in the nephron (independently of Na^+). Excessive ADH secretion leads to increased water reabsorption.* ECF volume increases; osmolarity of remaining ECF decreases. Water shifts from ECF to ICF.
- **Fluid shift:** From ECF to ICF
- **Compartments: ECF volume ↑, osmolarity↓, ICF volume ↑**
- **Blood findings: RBC volume ↑;** hematocrit approx. unchanged, **plasma protein concentration ↓**



Summary of Acute Fluid Shifts



Summary of Acute Fluid Shifts *(cont')*

Disturbances of Body Fluids

Type	Example	ECF Volume	ICF Volume	Osmolarity	Hematocrit	Plasma [Protein]
Isosmotic volume contraction	Diarrhea; burn	↓	N.C.	N.C.	↑	↑
Hyperosmotic volume contraction	Sweating; fever; diabetes insipidus	↓	↓	↑	N.C.	↑
Hyposmotic volume contraction	Adrenal insufficiency	↓	↑	↓	↑	↑
Isosmotic volume expansion	Infusion of isotonic NaCl	↑	N.C.	N.C.	↓	↓
Hyperosmotic volume expansion	High NaCl intake	↑	↓	↑	↓	↓
Hyposmotic volume expansion	SIADH	↑	↑	↓	N.C.	↓

ECF, Extracellular fluid; *ICF,* intracellular fluid; *NaCl,* sodium chloride; *N.C.,* no change; *SIADH,* syndrome of inappropriate antidiuretic hormone.

Costanzo Table 6.2

Fluid Shifts Caused by Chronic Renal Compensation

- **Long-term acid-base disorders** prompt the kidneys to slowly alter fluid, electrolyte, and acid-base retention to maintain homeostasis.
- Chronic compensation alters osmotic gradients, driving **gradual** fluid shifts.
- Fluid shifts caused by renal compensation are not represented by Darrow-Yannet diagrams and will be discussed later.

Feature	Renal Compensation	Acute Fluid Shift
Primary Trigger	Respiratory acid-base disturbances (CO_2 changes)	Osmotic or hydrostatic pressure gradients
System Involved	Renal tubule epithelial cells (proximal, intercalated)	Capillary endothelium and cell membranes
Timeline	**Slow** (Starts in 6–12 hours, peaks in 3–5 days)	**Immediate** (Seconds to minutes)
Mechanism	Altering HCO_3^- reabsorption and H^+ excretion	Bulk flow via Starling forces or aquaporins
Main Objective	Restore systemic pH toward 7.40	Restore volume or osmotic equilibrium

Questions?

